THIRD PARTY QUALITY ASSURANCE REPORT

of

Government Girls Degree College Rajpur Chajpur, Distt Muzaffarnagar (U.P.)

(Under Hon'ble Prime Minister's Janvikas Kariyakaram Yojna)

Submitted to

The Executive Engineer
U.P.R.N.S.S Construction Division
Saharanpur

by

Department of Civil Engineering
Z.H. College of Engineering and Technology
A.M.U. Aligarh
Aligarh

March 2021

80.31

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Certificate

We, hereby, certify that **Government Girls Degree College Rajpur Chajpur, Distt Muzaffarnagar (U.P.)** has been inspected personally by us and the matter presented in this third party quality assurance report is true to the best of our knowledge. The report cannot be used for any legal purpose.

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Acknowledgement

The investigators place on record their appreciation to the team members of **U.P.R.N.S.S Construction Division, Saharanpur** who extended all possible support for inspection of the work. The hospitality provided by them is extremely acknowledged.

The cooperation received from all the other officers and staff members is also gratefully acknowledged.

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1. INTRODUCTION AND GENERALOBSERVATIONS

The third party quality assurance inspection of **Government Girls Degree College Rajpur Chajpur, Distt Muzaffarnagar (U.P.).** has been undertaken for the third party quality assurance by Department of Civil Engineering, A.M.U., Aligarh, in reference to the task entrusted by the **Executive Engineer, U.P.R.N.S.S Construction Division, Saharanpur**, vide letter no 456/ U P R N SS/ NIPARAS dated 11.01.21.

The expert committee carried out visual inspection and the non-destructive tests of the Degree College. The observations made about quality of materials and workmanship relate to only what could be randomly seen at locations specified. As the work related to flooring, finishing, sanitary, water supply and electricity was not been started yet so it was not inspected. Engineers and supervisory staff shall thoroughly inspect the entire work for such defects as observed by expert committee as well as for other defects and take suitable remedial measures properly. The details of investigation, observations and comments are placed below.

S.No		Particulars of work	Comment
1.	(a)	Name of work	Construction of **Government Girls Degree College Rajpur Chajpur, Distt Muzaffarnagar (U.P.).** under Prime Minister's Janvikas Kariyakaram Yojna)
	(b)	Description of work	Civil Work, Sanitary Work, Electrical installations.
2.	(a)	Name of Senior Site Engineer Engineer	Shri Rohit Gautam
	(b)	Name of Executive Engineer	Shri Sandeep Kumar Indoria
	(c)	Name of Assistant Engineer	Shri Arvind Kumar
	(d)	Name of Junior Engineer	Shri Santosh Kumar

3.	(a)	Name of Agency	U.P.R.N.S.S Construction Divison, Saharanpur
	(b)	Contractor	M/s Malik Construction
	(c)	Registration class	A
	(d)	Agreement No	--
4.	(a)	Stipulated Date of start	29.06.2019
	(b)	Stipulated date of completion as per aggreement	28.06.2021
5.	(a)	Estimated cost	942.0 Lac
	(b)	Tender Amount	----
6.		Percentage progress at time of inspection vis a vis expected as per contract and reason for delay	20% of work is complete (only ground floor columns and tie beam casted till ceiling height of ground floor and infill masonry work completed till lintel level) Work delayed due to nonavailability of funds
7.		Officers and contractor present during inspection (Name and designation)	1. Shri Sandeep Kumar Indoria 2. Shri Santosh Kumar
8.		Date of inspection and number	24.03.2021
9.		Quality control aids used	1. Rebound Hammer 2. Digi bar locator
10.		Is site equipped with	
	(a)	Copy of agreement	Yes
	(b)	PWD Specification	Yes
	(c)	Is ISI marked/ approved materials used	Yes
	(d)	Guard file containing inspection reports of different agencies	Not at site; as informed all reports maintained by head office
	(e)	Testing facilities to check conformations to acceptance criteria	Yes , sample Reports attached Annexure I

11.		If fields laboratory existing and well equipped?	Not exist
12.		Have timely notices been issued to the contractor with the schedule of defects/damages and date of compliance? In case of failure to rectify defects/damages whether action initiated.	No notice issued
13.		Is soil investigation done?	Scope of this report is for super structure. However soil test was performed as informed by staff
14.	(a)	What is the source of water?	Submersible Pumps
	(b)	Has water been tested and approved by Engineer in-charge before construction?	No record available
	(c)	Has water been tested subsequently (i.e. after every 3 months) and found fit for use in works?	No record available
15.		Are all mandatory tests carried out at stipulated frequency for quality of materials?	Yes
16.		Specific control on RCC work like centering/shuttering, proportioning with boxes; mixing by full bag capacity hopper fed mixer; control of slump; placing/compaction with vibrator;	M20 concrete grade was used in tie beams and column. Since the RCC work was done before visit, so no comment can be given on the cantering, shuttering, cover block, line and alignment, levels etc. and the cleanliness of the formwork.
17.		Any other particular comments on the adequacy of process control.	Conducting regular quality inspection to ascertain quality of work.

18.		Observation on floors slope (especially in Bath, WC, etc.)	Floor slope not examined as flooring not done
19.		If dampness/leakages noticed, then state locations and probable reasons.	No dampness/leakages noticed
20.		Are sanitary fittings in place and functional?	Sanitary fittings not installed
21.		Are electrical fittings in place and functional?	Electrical work not started
22.		Were Samples collected by QC Core on site material?	Samples were tested in different laboratory and test report annexed (Annexure I)
23.		Observations on overall workmanship	Satisfactory
24.	(a)	Strength of concrete used in columns, beams, slabs, etc.	Refer test results for details
	(b)	Size and Strength of Foundation	It is buried, concealed part of structure were not possible for visual inspection.
25.		Fire Fighting Work position and lift provision	NA
26.		Quality of Water and Sanitary work	Not done
27.		Age of structure at the time of inspection	Construction work incomplete and on hold

2. NON-DESTRUCTIVE TESTS (NDT)

After visual inspection of construction of framed structure of the building of **Government Inter College Rajpur Chajpur, Distt Muzaffarnagar (U.P.)** testing using non-destructive tests (NDT) was started. It is observed through Digi cover that four 20 mm diameter steel bar on corners and four 16mm at center are used , cover provided is sufficient.

The non-destructive tests (NDT) are now being considered powerful methods of evaluating the existing concrete structures with regard to their strength and durability. The use of NDT has gained momentum over the last few decades owing to the ease of operation. It can very well supplement the other findings for a reasonably good estimation of the strength of a deteriorated concrete in a distressed structure. The ultrasonic pulse velocity (UPV) test and Schmidt rebound hammer (SRHT) tests are commonly used to determine the in-place quality and strength of concrete.

SRH test provide an inexpensive, simple and quick method of obtaining an indication of concrete strength. SRHT is based on the principle that the rebound of an elastic mass striking at the concrete surface depends on the hardness of the surface. On pressing the plunger of the rebound hammer against the concrete surface and then releasing, the spring pulled mass rebounds. The magnitude of rebound is recorded as RN. RN is a measure of energy absorbed during the impact thus an indicator of concrete strength. RH test only provides an estimate of the strength of concrete near the surface. RH test is most likely influenced by the surface layer of concrete and therefore considered as an 'indicator' of concrete strength.

2.1 Schmidt Rebound Hammer Test (SRHT)

SRH tests were performed on columns, and beams of framed structure of the college. The frequency and locations of the tests were randomly selected out on members. Table 1 provides a summary of SRH test results ie compressive strength of different components corresponding to their rebound numbers

Table 1:Strength Evaluation by Schmidt Rebound Hammer Test (SRHT)

Test No	Location	Type of Element	Position	Rebound Number	Mean Rebound	Compressive Strength (MPa)
1.	Porch	C1	Horizontal	28 30 28 28	28.5	22.33
		C2	Horizontal	26 28 30 30	29	23.00
2	Corridor	C4	Horizontal	28 30 30 30	29.5	24.22
		C6	Horizontal	28 30 36 30	31	25.88
		C10	Horizontal	30 30 32 28	30	25

		C13	I rizontal	30 32 30 32	31	25.88
		C15	Horizontal	30 32 30 30	30..5	24.55
		C16	Horizontal	28 28 30 30	29	23.00
3.	Rooms	C20	Horizontal	34 34 28 28	31	25.88
		C22	Horizontal	30 30 32 28	30	25
		C25	Horizontal	32 36 28 30	31.5	26
		C29	Horizontal	34 32 30 32	30	25
		C30	Horizontal	28 28 30 30	29	23
		C32	Horizontal	28 34 26 26	28.5	22.33
		TB1	Horizontal	30 30 30 30	30	25
		TB2	Horizontal	28 28 30 30	29	23
		TB3	Horizontal	26 28 30 30	29	23
		TB4	Horizontal	30 28 33 29	30	25

NOTE: C stands for Column; TB stands for Tie Beam

3. CONCLUSIONS

1. 20% of construction work is complete (Only columns till roof level of of Ground floor and tie beam casted, Masonry work complete till lintel level)
2. Physical observations show no damage or fault in the structural components built at this site and therefore construction appears to be safe and structurally sound.
3. The quality of civil construction works completed till date of visit which were tested by non-destructive method were found to be satisfactory.
4. Average strength of concrete used was found as 24.28 MPa.
5. The conditions of foundations was not possible for visual inspection. Hence, could not be considered for inspection.
6. The bricks used for the walls were of fair quality, plastering not done yet..
7. The engineers and contractor were given verbal tips how to maintain quality during construction.

8. The agreement has been checked out and found that completion of the work is delayed.

9. This is only the expert opinion based on certain tests performed at sites and certain statistical data analysed to verify the test results. The results are found to be satisfactory. This report cannot be used for any legal proceedings.

10. Some of the photographs taken from the site during inspection /testing is attached at Annexure I.

11. The total report comprises of twelve pages excluding photographs and Annexures.

4.REFERENCES:

1. Concrete Microstructure, Properties and Materials- P.K. Mehta Paulo and J.M. Monterio, 1999.

2. Guide to Durable Concrete - ACI Committee 201 Report, ACI Journal, Dec. 1977.

3. IS:456 (2000)- Code of Practice for Plain and Reinforced Concrete, BIS, New Delhi.

4. IS:13311 (Part 1):1992 Non-destructive Testing of Concrete – Methods of Test [Part 1 Ultrasonic Pulse Velocity]

5. IS:13311 (Part 2):1992 Non-destructive Testing of Concrete – Methods of Test [Part 2 Rebound Hammer Test]

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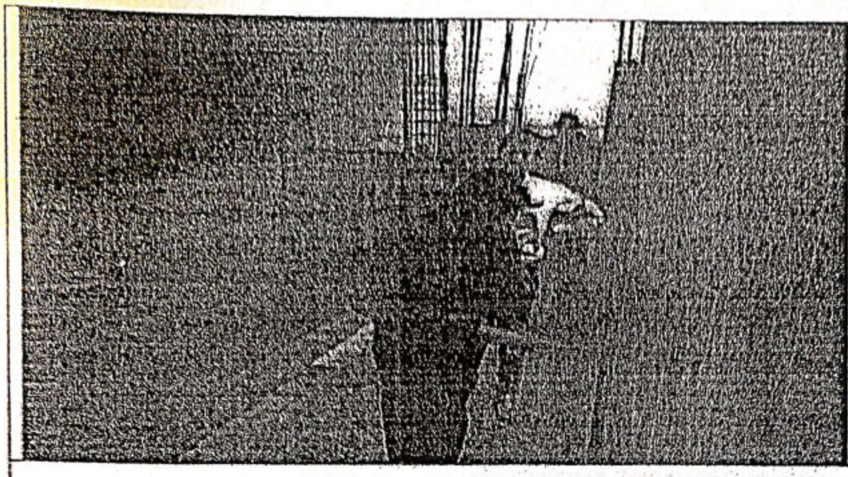
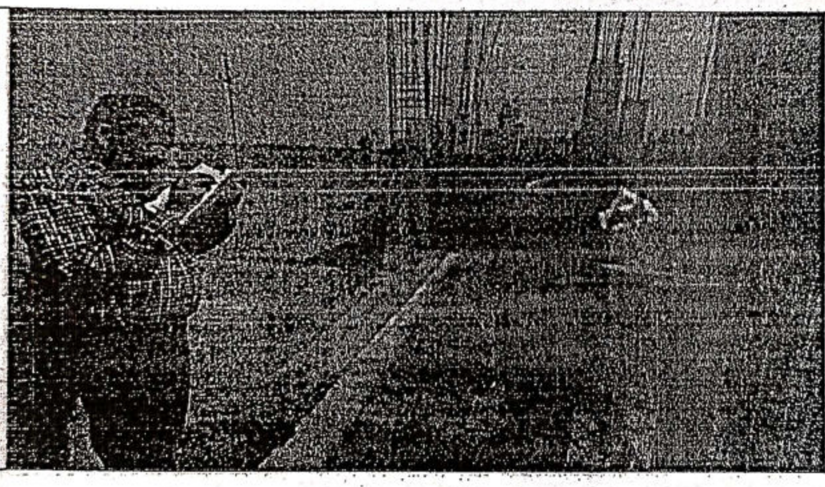
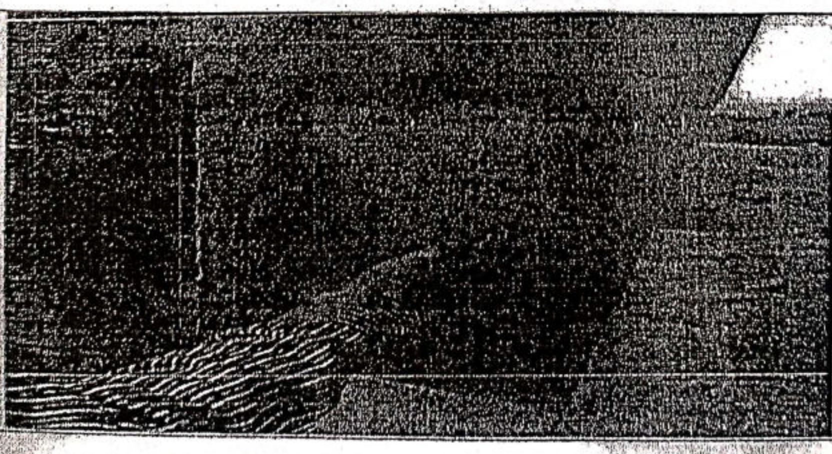
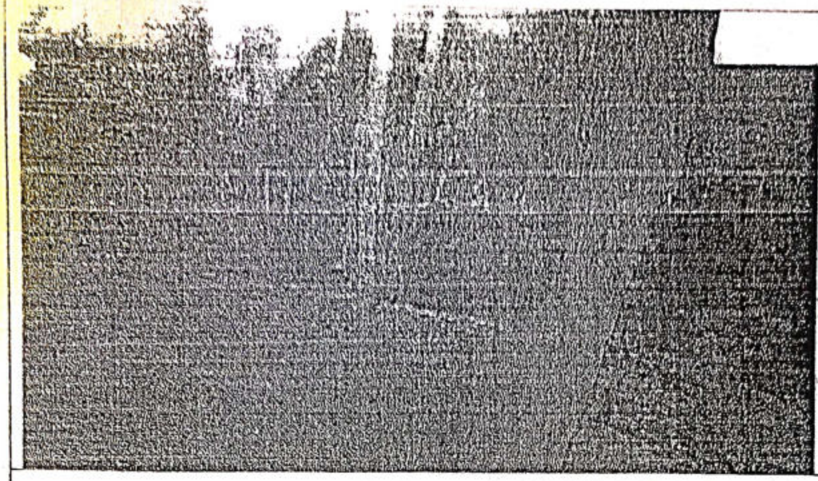
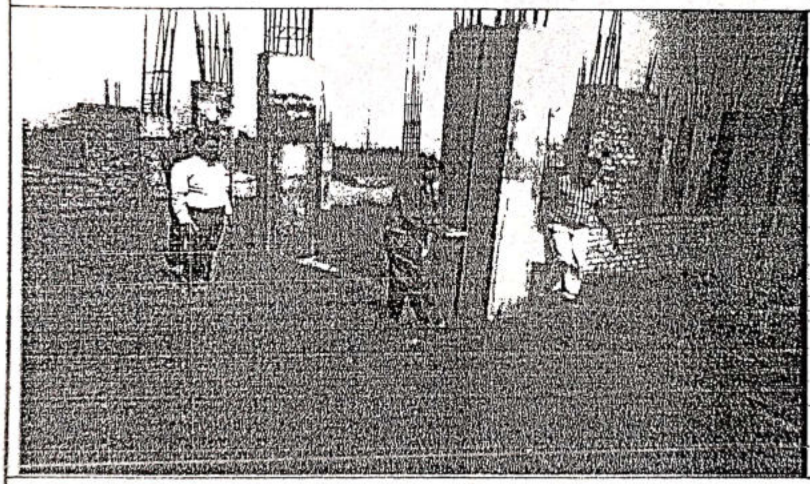
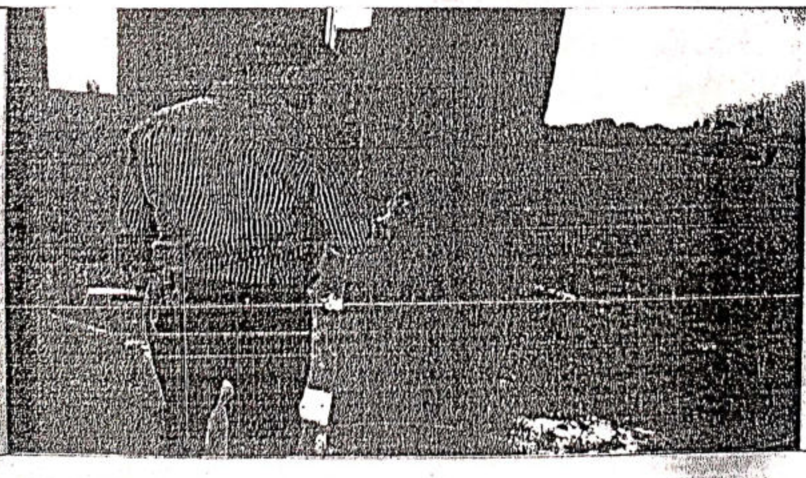
	TESTING OF COLUMN USING REBOUND TEST
	TESTING OF COLUMN USING REBOUND TEST
	TESTING OF COLUMN USING REBOUND TEST

	TESTING OF TIE BEAM USING REBOUND TEST
	TESTING OF COLUMN USING REBOUND TEST
	TESTING OF COLUMN USING REBOUND TEST

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CIVIL ENGG. DEPTT. CONSULTANCY DIVISION TESTING REPORT

Ref No. 1571/Civil Engg./2022-23 Date 02/01/2023

NAME OF DEPARTMENT कार्यालय - अधिशासी अभियन्ता उत्तर प्रदेश राज्य निर्माण सहकारी संघ लि० दिल्ली रोड सहारनपुर।

NAME OF THE TEST Compressive strength of Mix Ratio (1:1½:3) M-20 Cement Concrete Cube after 28 days Curing.

REF. NO. Memo/नि.प्र.स./2022-23 .... दिनांक - 21/11/2022

Cube Cast date - 21/11/2022

S. No.	NAME OF SPEC.	NAME OF TEST	RESULT	SITE DETAILS
1.	15×15×15 c.m³ c.c Cube	Comp. strength	203.18 Kg/cm^2	राजकीय कन्या डिग्री कालेज रामपुर रामपुर जनपद मुजफ्फरनगर
2.	— do —	— do —	205.41 Kg/cm^2	के IInd floor slab &
3.	— do —	— do —	201.12 Kg/cm^2	beam Casting का निर्माण कार्य।
4.				
5.				
6.				

Note 1- Sample Collected by your Agency.
2- Site details as per your letter.

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Ref No. 1571/Civil Engg./2022-23 Date 02/01/2023

NAME OF DEPARTMENT कार्यालय अधिशासी अभियन्ता उत्तर प्रदेश राज्य निर्माण सहकारी संघ लि० दिल्ली रोड सहारनपुर।

NAME OF THE TEST Compressive strength of Mix Ratio ($1:1\frac{1}{2}:3$) M-20 Cement Concrete Cube after 28 days Curing.

REF. NO. Memo/नि.यु.स./2022-23 .... दिनांक-21/11/2022

Cube Cast date-21/11/2022

S. No.	NAME OF SPEC.	NAME OF TEST	RESULT	SITE DETAILS
1.	15x15x15 cm C.C Cube	Comp. strength	203.18 Kg/cm²	राजकीय कन्या डिग्री कालेज रामपुर हसनपुर जनपद मुजफ्फरनगर के IIIrd Floor slab & beam casting का निर्माण कार्य।
2.	—do—	—do—	205.41 Kg/cm²	
3.	—do—	—do—	201.12 Kg/cm²	
4.				
5.				
6.				

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2- Site details as per your letter.

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CIVIL ENGG. DEPTT. CONSULTANCY DIVISION TESTING REPORT

Ref.No. 1570/Civil Engg/2022-23 Date. 02/01/2023

NAME OF DEPARTMENT कार्यालय - अधिशासी अभियन्ता उ०प्र० प्रदेश राज्य निर्माण सहकारी संघ लि० दिल्ली रोड सहारनपुर।

NAME OF THE TEST Compressive strength of Mixed Ratio (1:$1\frac{1}{2}$:3) M-20 Cement Concrete Cube after 28 days Curing.

REF. NO. Memo/नि.प्र.स./2022-23 दिनांक - 07/11/2022

Cube Cast date - 07/11/2022

S. No.	NAME OF SPEC.	NAME OF TEST	RESULT	SITE DETAILS
1.	15x15x15 c.m³ c.c Cube	Comp. strength	200.90 kg/cm²	राजकीय कन्या डिग्री कॉलेज रामपुर मनिहारान
2.	— do —	— do —	202.85 kg/cm²	जनपद - मुजफ्फरनगर के Second floor slab
3.	— do —	— do —	201.58 kg/cm²	व beam casting का निर्माण कार्य
4.				
5.				
6.				

Note 1- Sample collected by your Agency.
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CIVIL ENGG. DEPTT. CONSULTANCY DIVISION TESTING REPORT

Ref No. 1579/Civil Engg/2022-23 Date. 02/01/2023

NAME OF DEPARTMENT कार्यालय - अधिशासी अभियन्ता उत्तर प्रदेश राज्य निर्माण सहकारी संघ लि० दिल्ली रोड सहारनपुर ।

NAME OF THE TEST Compressive strength of Mixed Ratio (1:1½:3) M-20 Cement Concrete Cube after 28 days Curing.

REF. NO. Memo/नि.उ.स./2022-23 दिनांक - 07/11/2022

Cube Cast date - 07/11/2022

S. No.	NAME OF SPEC.	NAME OF TEST	RESULT	SITE DETAILS
1.	15x15x15 cm³ C.C Cube	Comp. strength	200.90 kg	राजकीय कन्या डिग्री कॉलेज रायपुर हसनपुर
2.	— do —	— do —	202.85 kg/cm²	जनपद - मुजफ्फरनगर
3.	— do —	— do —	201.58 kg/cm²	के second floor slab & beam casting का
4.				निर्माण कार्य
5.				
6.				

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